Objects and their images in a lens

2a

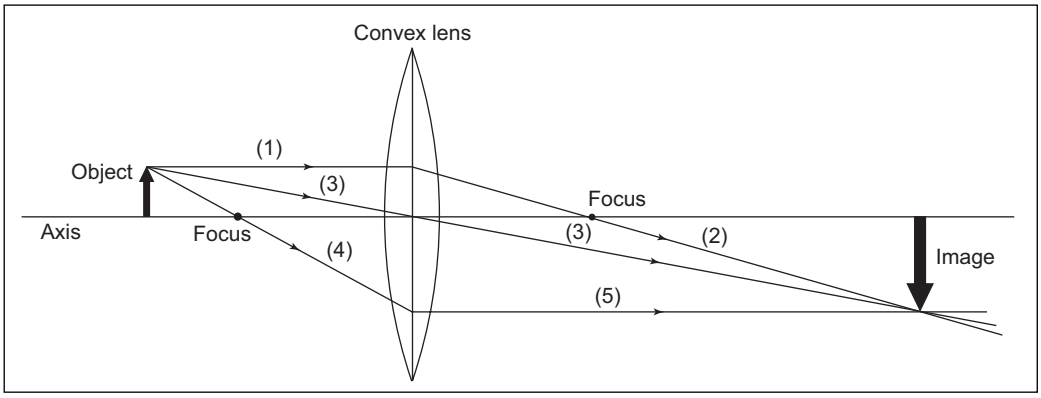
If you want to see an object clearly when using a lens, you have to hold the lens a certain distance away from the object and a certain distance from your eye. It is possible to predict where the focused image will be, if the object is a known distance from the lens. Of course when using a lens, you just move your eye to where the image is clearest and sharpest. To position images by drawing, a couple of rays from a point on an object (e.g. the tip of an arrow) are selected, and drawn following some rules about refraction. The refracted image position of the arrow tip will be given by where these rays meet.

Here are some things about light rays that you need to know to draw images in a convex lens:

- A parallel ray is a ray that is parallel to the axis.
- A focus ray is a ray that travels through the focus.
- A centre ray is a ray that passes through the centre of the lens.
- An incoming ray travels from the object to the lens.
- A refracted ray travels away from the lens.
- Light rays should be drawn as though they refract (i.e. bend) at the middle of the lens.
- After an incoming parallel ray (1 in the diagram below) is refracted, it becomes a refracted focus ray (2).
- An incoming centre ray (3) continues in a straight line through the lens.
- After an incoming focus ray (4) is refracted, it becomes a refracted parallel ray (5).
- If incoming rays do not reach the lens, extend them backwards.
- If refracted rays do not seem to meet, extend them backwards.
- If refracted rays do not meet, a clear image is not produced.
- If the image is on the other side of the lens to the object, it is a real image.
- If the image is on the same side of the lens as the object, it is a virtual image.

Compared to the object, an image can be closer to the lens, further from or the same distance from the lens as the object; smaller than the object, larger or the same size; upright or inverted; and real or virtual.

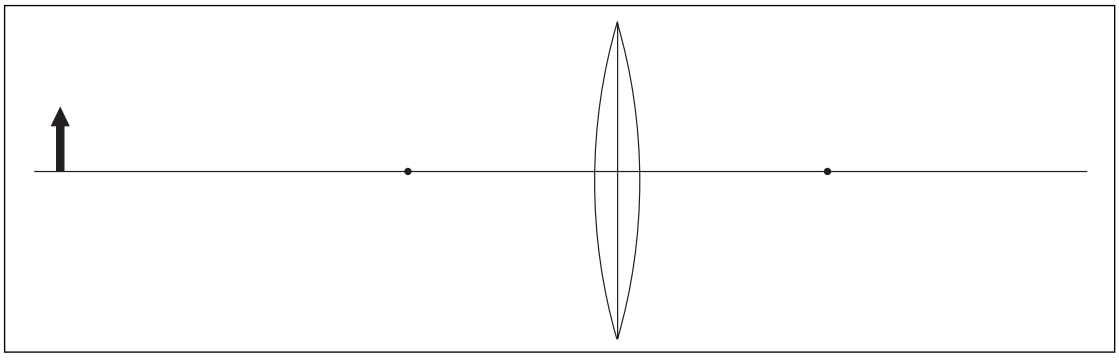
The diagram shows how the image for an object placed near a convex lens can be drawn.



The image is further from the lens, larger than the object, inverted and real.

1 Construct the image of the object in each of the following diagrams. Follow the rules listed above. Compare the image with its object.

a

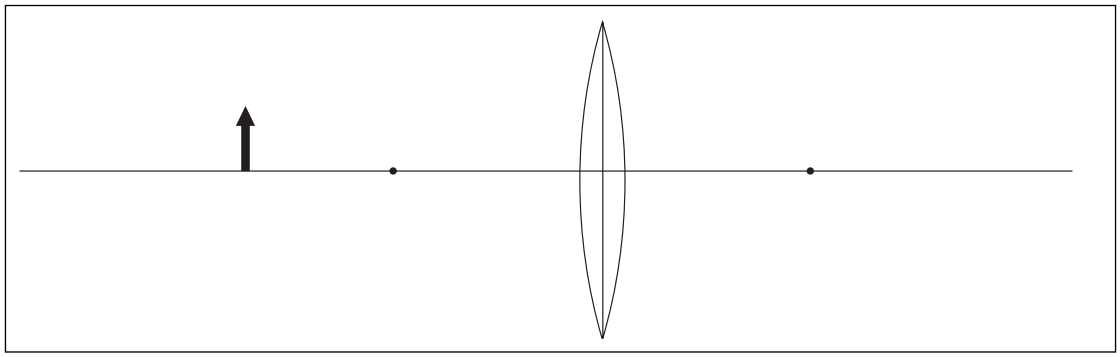


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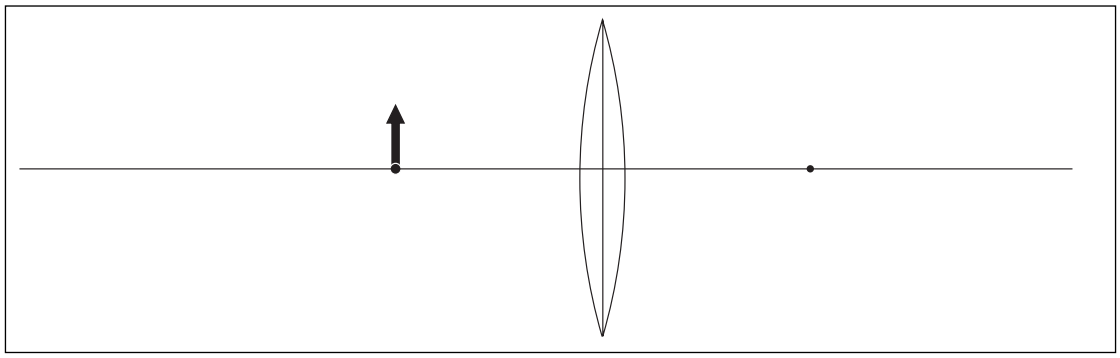
Objects and their images in a lens

2b

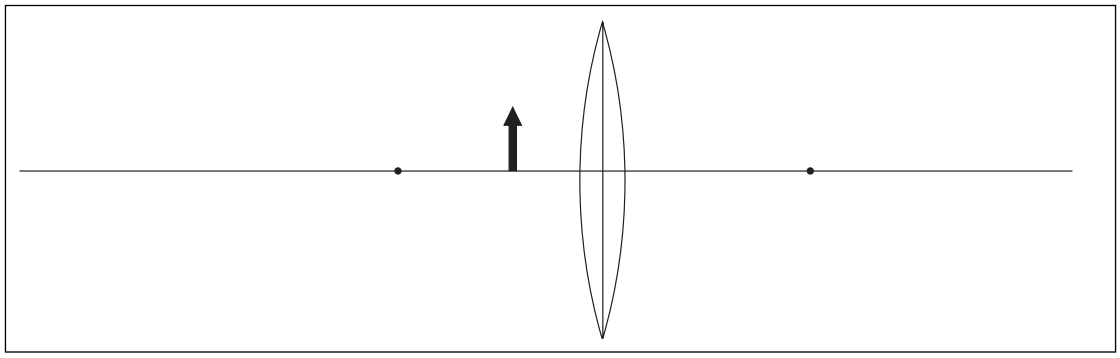
b



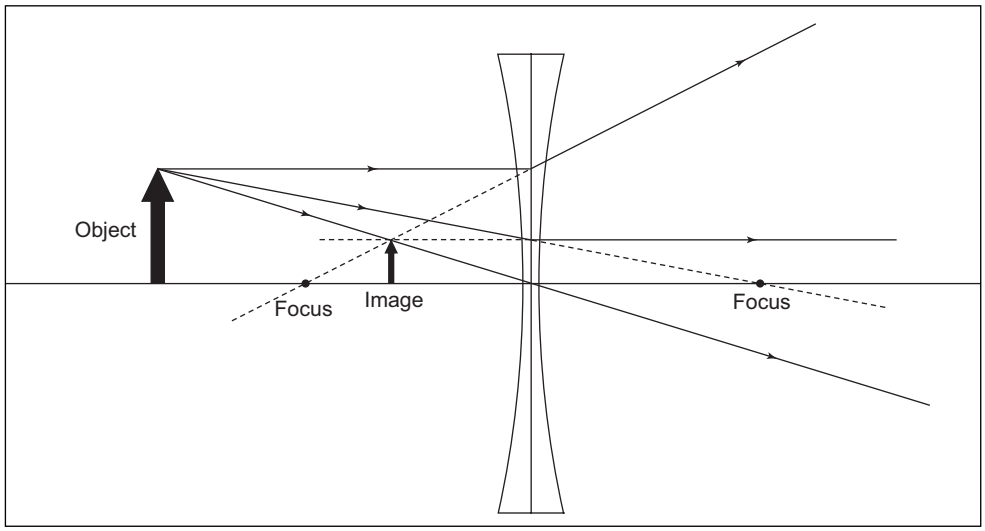
c



d



Below is an object–image diagram for a concave lens. The diagram shows the paths of three significant rays that are important in the drawing of the image.



The image is closer to the lens than is the object, smaller than the object, upright and virtual.

2 Study the diagram and develop a set of refraction rules that will help you draw images produced by concave lenses. Then construct a similar diagram with the object in a different position and apply your rules to find and describe the image. Do these tasks in your science book.

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